Course 1

- CAA means designing and building apps using services provided by cloud environments to achieve greater:
  - scalability
  - security
  - cost efficiency
  - availability
  - productivity
  - felexibility
  - speed
  - performance
- on - demand

Traditional Approach:

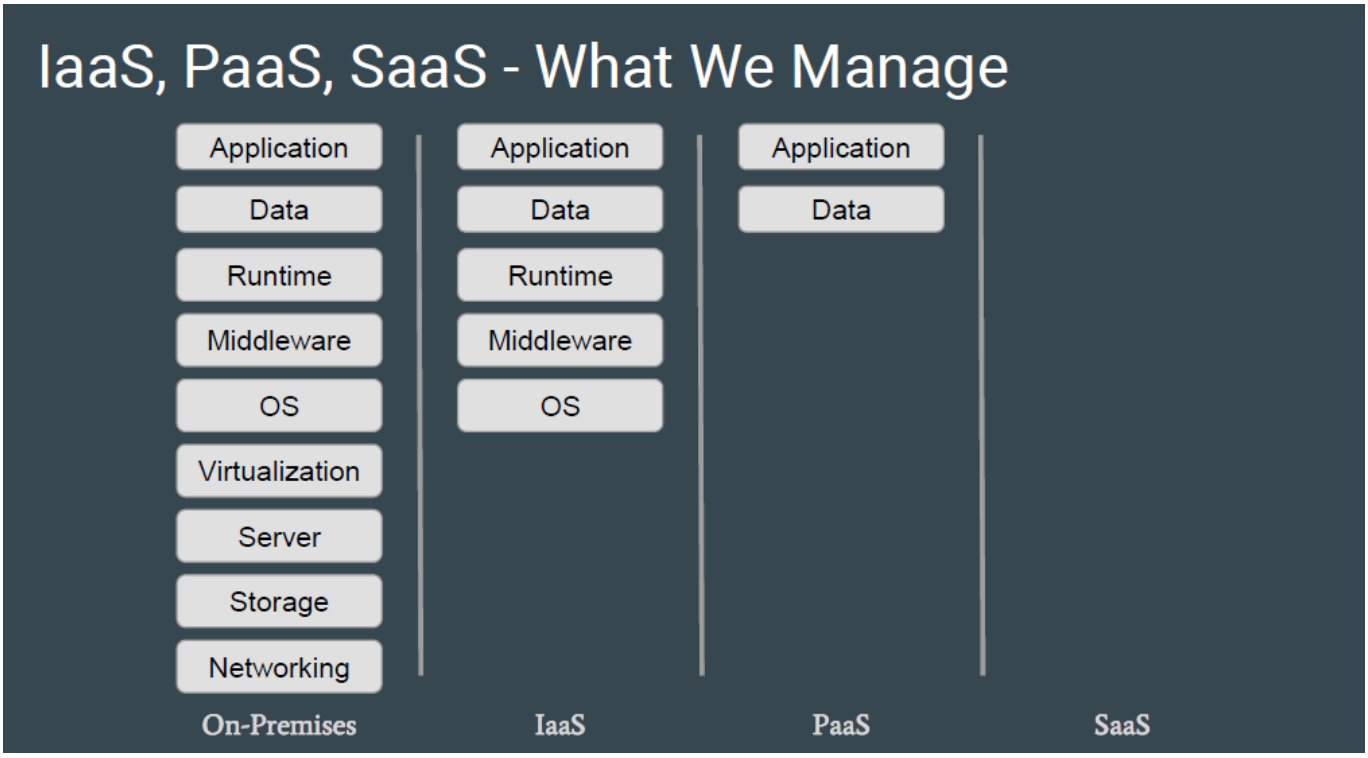
- On-premises data centers (Mono-lithic, hosting,rent server,install , copy app with FTP,run)
- For larger comp: buget, data center problems

Cloud Classification:

Deployment Models: - public -> AWS, Azure, Google Cloud - private -> dedicated - hybrid -> mix of public and private - community -> shared by several organizations

Service Models: - IaaS -> Infrastructure as a Service - PaaS -> Platform as a Service - SaaS -> Software as a Service BaaS,DBaaS,FaaS -> are PaaS

What is managed in each service model:



la IaaS doar:

```
- App
- Data
- Runtime
- Middleware
- O/S
```

usually vm services. Other services are built on top of IaaS. (Amazon EC2,Google Compute Engine, Azure Virtual Machines)

la PaaS doar:

```
- App
- Data
```

lets u develop ur solution without woring about how it runs.
E.g:

- Cloudfoundry
- Google App Engine
- Azure App Service
- RDS
- Cloud SQL

la SaaS nimic

- u just use the app eg: - Office 365
    - Dropbox
    - Slack

Standards/ Compliance:

```
- CSA (Cloud Security Alliance) - Best practices for cloud security
- ISO 9001 - Global Quality Standard
- ISO 27001 - Information Security Management
- ISO 27017 - 27001 in the context of cloud
- ISO 27018 - Personal data protection in cloud
- PCI DSS - Card payments security
- HIPAA - Health information protection
```

Pricing Models:

Time based

```
- second/hour when service online
    - IaaS and PaaS
- usually la vms, dbs and general PaaS services
- easy to control costs
```

Usage based

- pay per unit - mostly SaaS or Paas
- de obicei la fully managed services (Faas,DBaas-NoSQL usually) si storage
- pay based on trafic
- nu poti modifica costul fara sa opresti serviciul

Tipuri

Prices can be fixed,tiered and or dynamic. OR SUBSCRIPTION BASED.

Fixed:

- same price for all usage
- de obicei on demand services

Tiered:

- less per unit the more you use
- de obicei la storage and bandwidth

Dynamic:

- based on supply and demand
- mai mult spatiu neutilizat, mai ieftin

Arhitecture Framerworks:

- Operational Excellence
    - Automation, monitoring, change management
- Security
    - IAM, defense in depth, traceability
- Reliability
- Performance
    - scaling,design
- Cost Optimization
    - scaling,understand services,monitoring

Course 2

Iaas

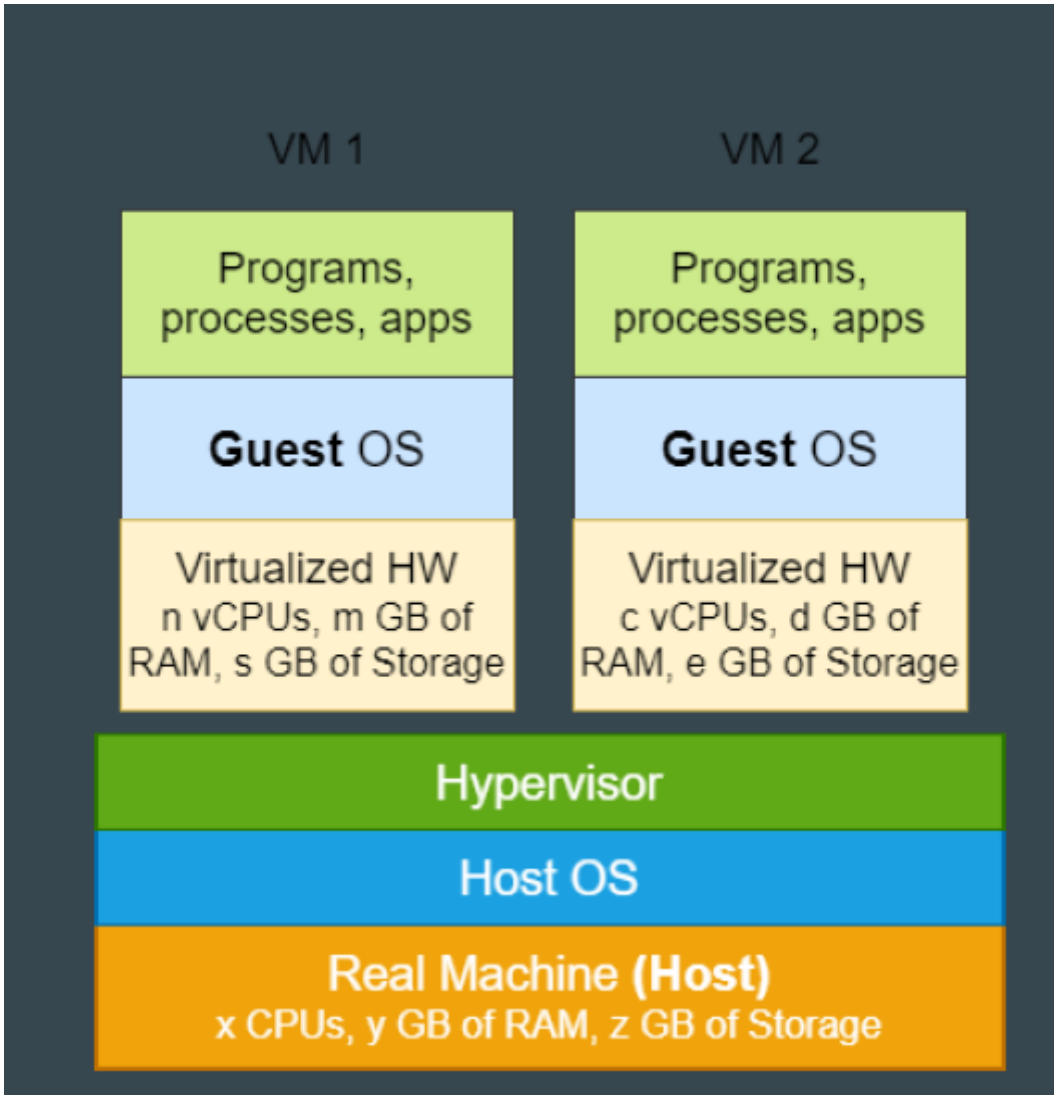
- application
- data
- runntime
- middleware
- os

Ce este?

- use virtualized infrastructure on demand without having to worry about the underlying hardware

VMs

- emulated computers
- host machine where the vm runs
- guest - vm
- hypervisor - software that runs vms



HyperVisor

- optimize resource usage
- multi tenant usage (vms for multiple clients) - ensure isolation, optimize cost
- most providers have custom (AWS - Ntro, Azure - Hyper-V, Google - KVM)

- or support others (VmWare)
- u choose the config
- general purpopse , cpu , memory , gpu
- some have pre config options

VCPU

- pot fi timeshares o real cpu, pyshical core, or logical core(hyperthreading)

VM - Remote Access

- SSH - Linux
- RDP - Windows
- security concern
- accesate prin bastion host - interm vm

Monitoring

- cpu
- disk
- network
- memory App: CloudWatch, Azure Monitor, Google Stackdriver ELK stack - ElasticSearch, Logstash, Kibana

Exemple VM

- Amazon EC2
- azure vm
- gce
- droplets

Automation

- **Infrastructure as Code** - automate infrastructure deployment and management
- in AWS: CloudFormation, Terraform
- in Azure: ARM templates
- in Google: Deployment Manager Images:
- manually config a VM, create an image, spin up identical vms
- best practice: use images to create vms

Dedicated Services:

- ansible, chef, puppet for automation config infract
- some require a master node

Storage

- usually separate from vms
- volumes can be attached to different vms Block Storage (as opposed to object storage) Boot Disk - fisier butare sistem SSD, HDD,Magnetic Encryption Snapshots - backup Instance store - temporary storage NAS - Network Attached Storage

Networking

- netword cards are attached to vms

Workflow: Working with IaaS

1. Create a VM using one of the tools made available by the provider a.Web console b.CLI c.SDK d.API
2. Connect to it (optional)
3. Install the necessary programs (can and should be automatized) a.Application dependencies b.Monitoring software
4. Deploy your application (can and should be automatized)
5. Monitor the VM and app
6. Keep the VM (OS + programs) and app up-to-date

When to use IaaS

- first migration (lift and shift) - move existing apps to cloud
- special programs - native apps - need specific OS
- ultra high performance - u can rent physical servers
- special licenses - need to run on specific OS
- trying to avoid vendor lock-in (rarely worth) - can move to another provider
- long running workloads- cheaper than alt
- want to have more control over the infrastructure

Course 3 - Networking in the Cloud

Adrese private: 10.0.0.0- 10.255.255.255 172.16.0.0 - 172.31.255.255 192.168.0.0- 1...

Routing

- process of selecting the best path to a destination Protoocls: - IP - Internet Protocol - BGP - Border Gateway Protocol - OSPF - Open Shortest Path First - RIP - Routing Information Protocol Delivery schemes: - Unicast - one to one - Broadcast - one to all - Multicast - one to many - Anycast - one to nearest

NAT - Network Address Translation

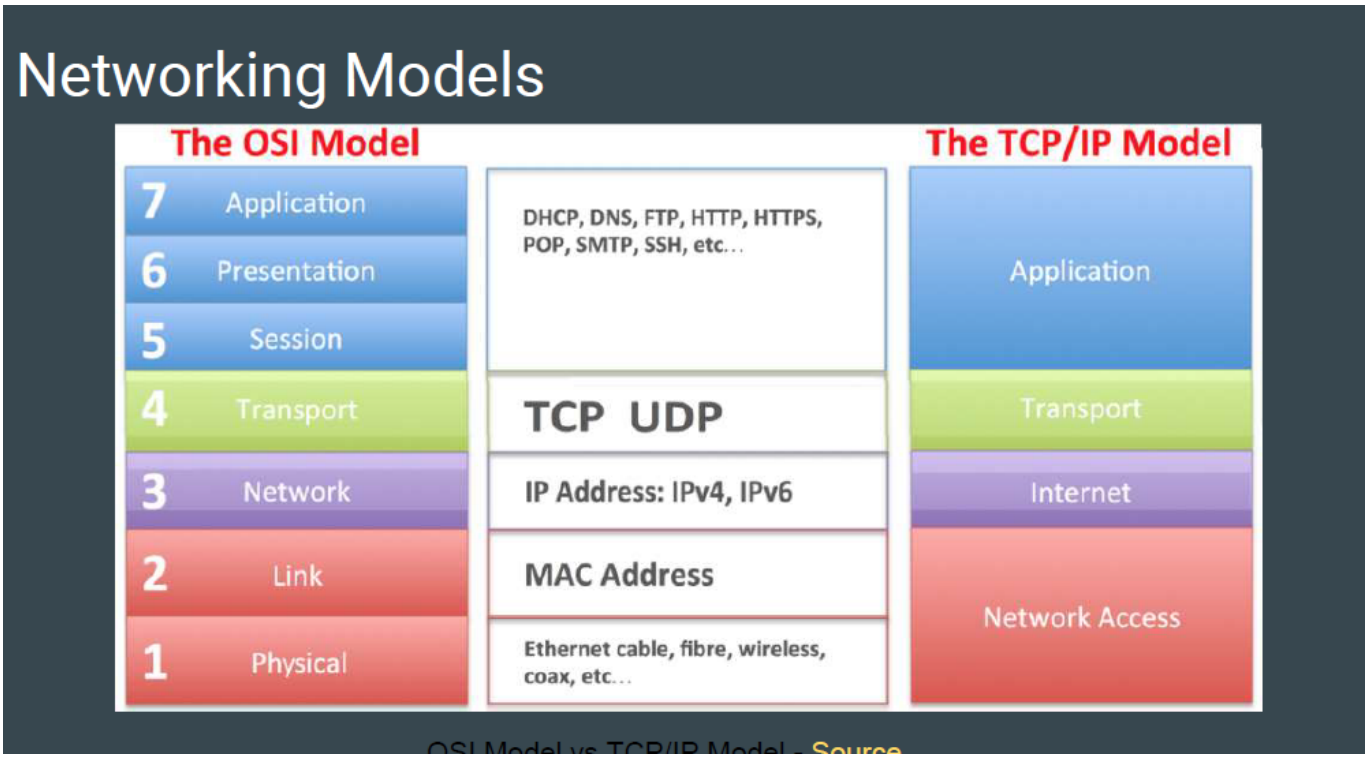
- allows multiple devices to share a single public IP
- pot translata si porturile

Gateway

- device that connects two networks

- networks usually have default gateways

Networking models



TCP/IP model

- Application : dhcp, dns, http,pop,smtp
- Transport : tcp, udp
- Internet : ipvp4, ipv6, icmp
- network acces: - link layer: mac address - physical layer: cables, switches, routers OSI:
- Application (7) - app
- Presentation (6) - app
- Session (5) - app
- Transport (4) - tcp, udp - transport
- Network (3) - ip - internet
- Data Link (2) - mac - network access
- Physical (1) - cables, switches, routers - network access

Cloud Networking

- cel mai relevand in IaaS
- constantly evolving (VPC 2020)
- dependend on provider

VPC - Virtual Private Cloud

- isolated network in the cloud
- works with private ip addresses
- can be split into subnets

Components

- network interfaces
- route tables
- gateways
- nat gateways
- dns

Peering

- allows two vpcs to communicate
- the responsability to avoid handle ip overlaps is up to us
- between on prem and cloud

Common vulnerabilities:

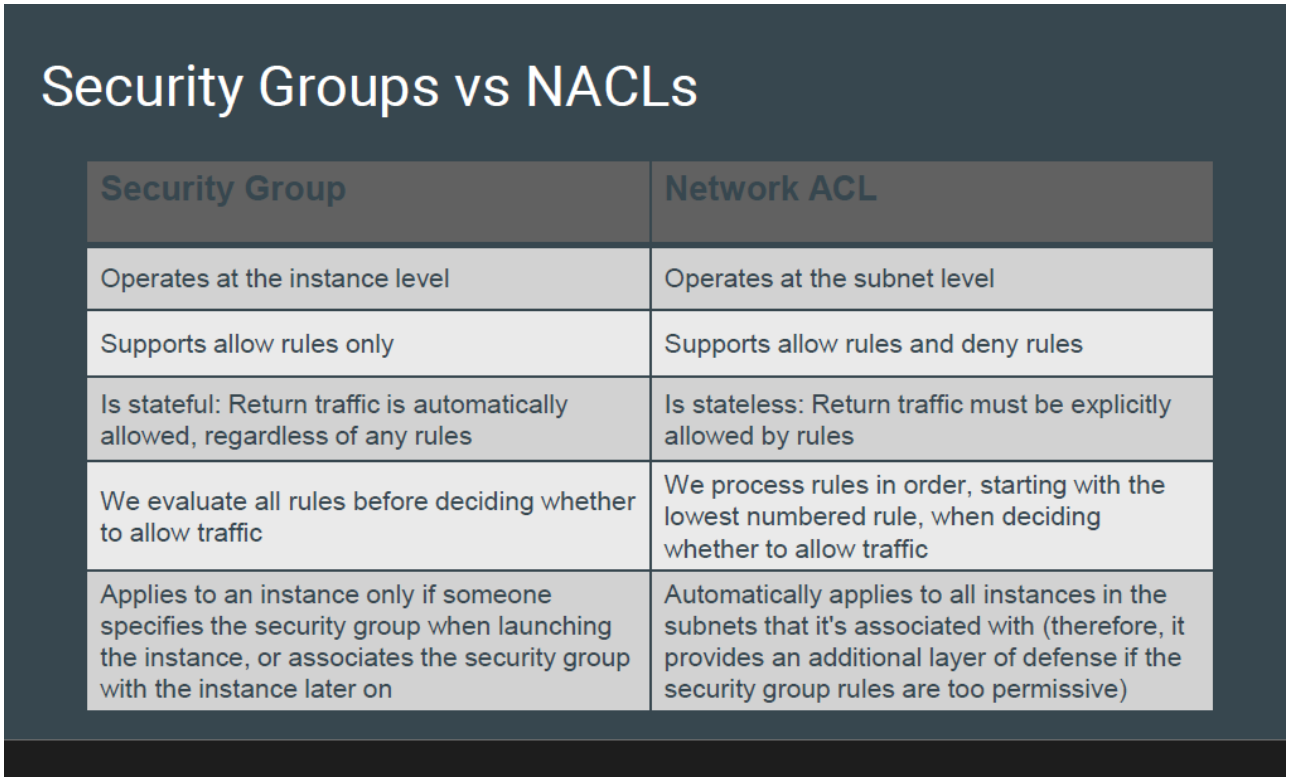
- we cannot choose to deploy the most mangaded services la un moment dat

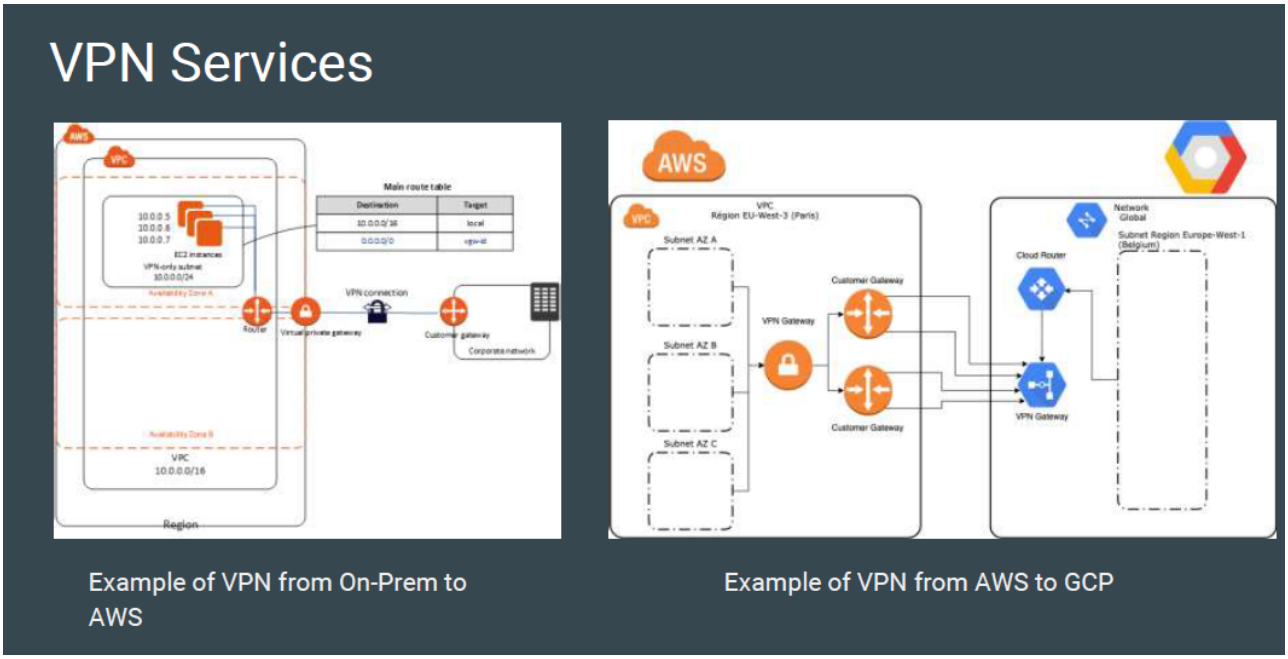
Network Performance:

- bigger vms -> most likely better v network cards
- some support clustering
- packets should enter the provider's network as soon as possible

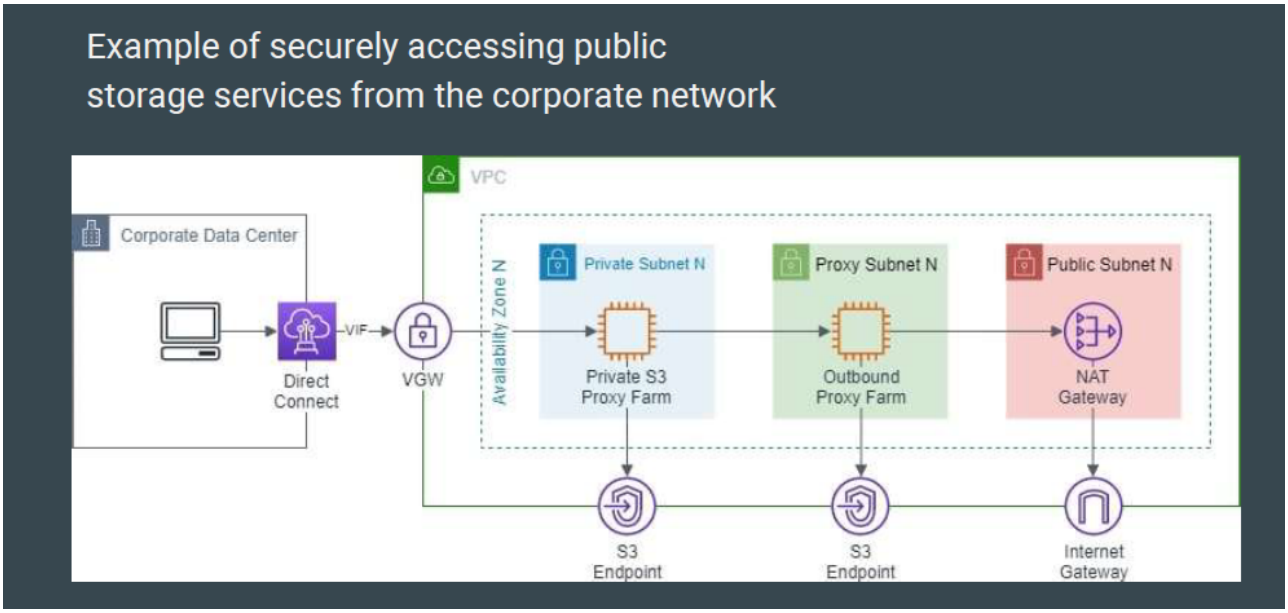
Security groups:

- act as a virtual firewall Network Access Control Lists:
- stateless (not like firewalls that are stateful)





Direct Connection Services When VPN is not enough (usually performance-wise). Is a physical connection between customer network and cloud provider. Highest performance achieved when connecting directly to the cloud infrastructure, but partners also offer this (enough in most cases) (e.g. Orange and Vodafone).

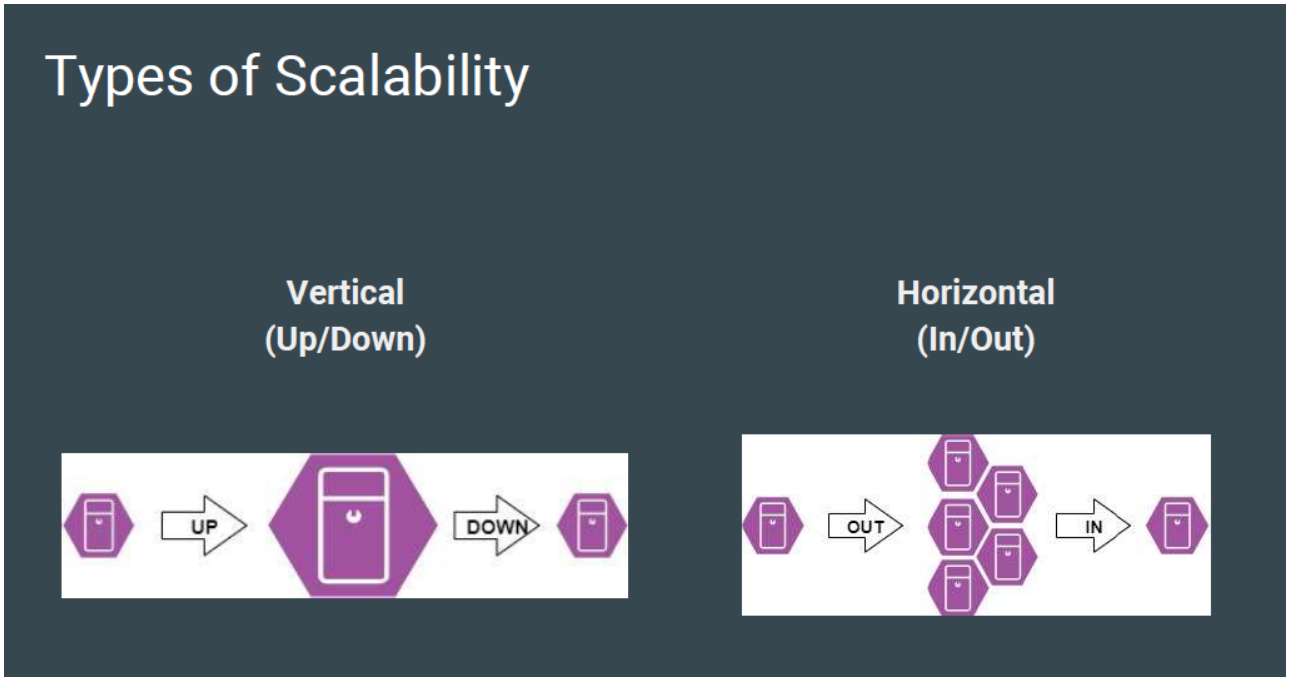


Course 4 - Scalability

Tipuri de scalabilitate:

- Vertical - mai multe resurse pe un singur server

- Horizontal - mai multe servere



Comparison Vertical ● Usually easier/straightforward to implement (update the VM). ● Compatible with any app architecture (e.g. stateful apps). ● Doesn't require additional components. ● Usually requires restarting the service. ● Might be the only approach for certain components (e.g. RDBMS) Horizontal ● Infinite scale. ● Requires stateless apps (promotes better architectures). ● Requires load balancers (not always). ● Does not touch already existing VMs (i.e. no downtime) (unless when scaling in).

Horizontal Scalability

- posible unlimited scale
- no restart

Scaling strategies

- contorizare cereri, memorie, cpu
- aim to keep load under 50 buffer pana pornesc alte vms
- monitorizare delay
- dureaza ceva scalarea
- initializare,etc
- s ar putea la load balancer sa ii ia ceva sa foloseasca noua vm Mai poti face si schedule scaling
- ajuta sa ai resources pre warmed

Traffic Routing on horizontal scaling: - load balancer in front of vms : un singur punct de intrare - dns load balancing - app aware routing

Load Balancers

- balance traffic between vms
- monitorizeaza vms
- single point of entry
- some caching
- secuirty: SSL offloading, authentication,A/B testing

- utilizare websockets,http,https
- auto scaling
- routing based on path,host,query string
- health checks
- integration with other services
- cross region load balancing

Types:

Layer 7 - HTTP

Works at layer 7 of the OSI model

- Offers most features
- Adds significant latency (~400ms)
- Behaves as a Proxy (Sends new requests)
- Good default choice

Layer 4 - TCP

Works at layer 4 of the OSI model - i.e. works with packets

- Behaves similar to a network router (the request is not decomposed)
- Ultra-high performance
- Not so many features

DNS - layer 7:

Also works at layer 7

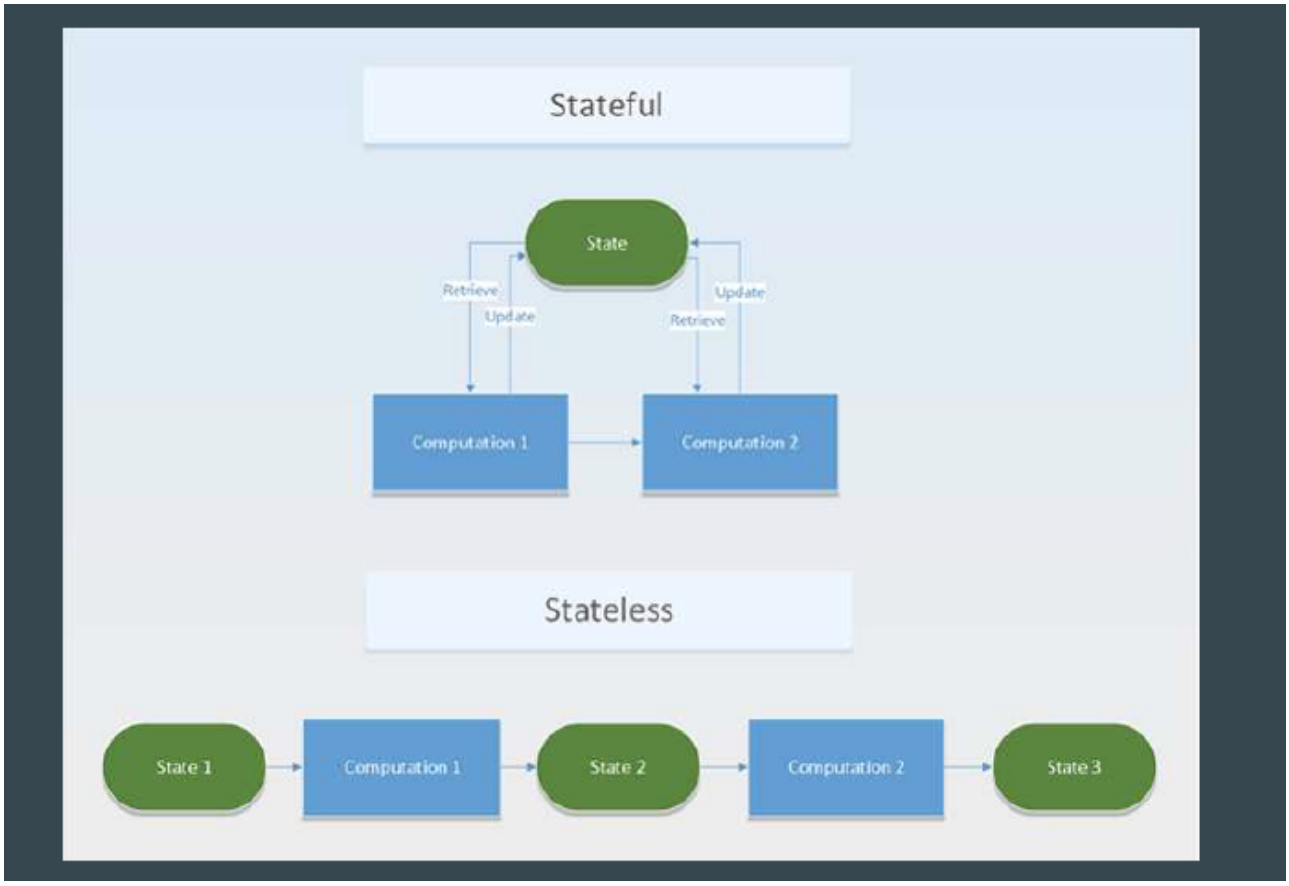
- Useful for global load balancing
- A domain might be pointing to several IP addresses. Clients usually connect to the first one - send IPs in desired order.
- Issues with caching/TTL

Routing Strategies: Round-Robin ● Simplest, route traffic in a loop. Good enough for simple use cases Weighted Round-Robin ● Prefer certain targets more than others (useful when target vary in sizes) Least Connections ● Useful when requests vary in load

Application Layer Scaling

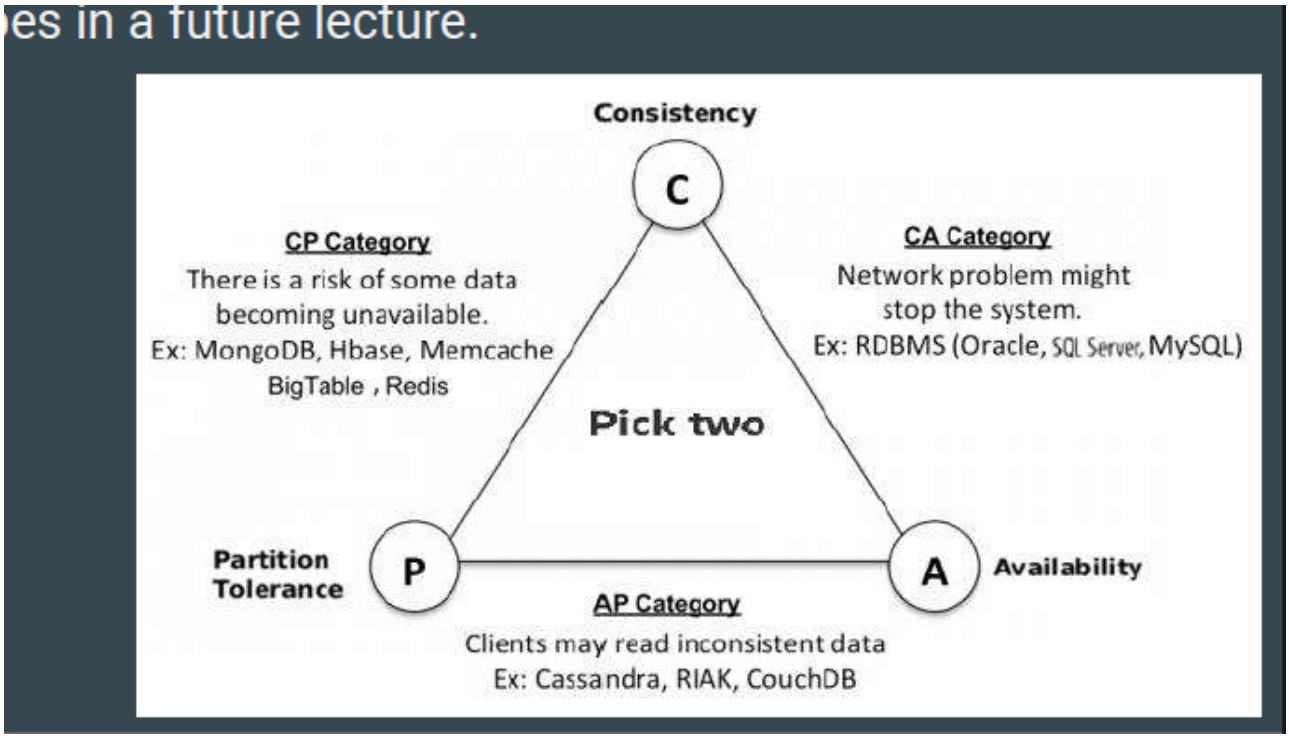
Stateless

- fiecare request independent



Database Scaling

- nosql scale easier



Consistency Types

- se refera la cat de repede se propaga o schimbare in baza de date

Strong

- toate citirile dupa o scriere vor vedea schimbarea

Eventual

- schimbarea se va propaga in timp
- acid vs base se paoate face scale pe reads multiple nodes for read

Partitioning

- split data in multiple nodes

Sharding

- share data in multiple nodes
- shard key
- distribuirea datelor pe mai multe noduri

Course 5 - Web Capabilities

URI - Uniform Resource Identifier URL - Uniform Resource Locator - uri specifica locatia URN - Uniform Resource Name - uri specifica numele

HTTP

HTTP - Hypertext Transfer Protocol

- how messages are formatted and transmitted
- layer 7 of OSI model
- stateless (nu ca TCP)
- common actors: client, server, proxy

Versions

1.0 - intial need for tcp connection for each request 1.1 - persistent connections, pipelining, chunked transfer encoding 2.0 - derivat from SPDY -> server push 3.0 - works over udp

Headers

Content-Type - type of data CacheControl - how long to cache Authorization - credentials Access-Control-Allow-Origin - CORS Origin - where request comes from Header fields:

- general : request url, request method, status code, remote address
- request: cookies,accept -xx, content type,length,authorization,user-agent
- response : server,set cookie, content type, length, location

Content storage

- S3 - simple storage service Amazon

- Cloud Storage Google
- Azure Storage Account(Blobs,Files,Queues,Tables)

Caching

- mai multe nivele poate sa se afle: - server - browser - cdn cdn si browser usually controled he Header Cache-Control si content static si dinamic poate fi cacheuit dar cu TTL diferit

Distribution

- latenta mare la distanta mare
- paid content : signed requests

Website Hosting

Static

```
- no processing - html,css,js
- cheap - S3, Cloud Storage
```

Dynamic

```
- server with cpu
```

Content Delivery Networks - CDN

- cache content closer to the user (EDGE CACHING)
- improve load time (content closer)
- optimize cost (content heavy apps, traffic costs)
- reduce presure on critical resources
- improve security (DDOS protection) eg: CloudFront, Akamai, Cloudflare

DNS

- translate domain names to ip
- multiple layers DNS Actors: DNS Recursor ● Root nameserver ● TLD nameserver ● Authoritative nameserver Nameserver- > dns server handling requests Records : domain to server CNAME - alias MX - mail server TXT - text eg: Route 53, Google Cloud DNS, Azure DNS,Cloudflare DNS

DNS TTL - Time to Live - how long to cache

Course 6 - High Availability

Ce contribuite la HA:

- hardware
- software
- network
- data

formula: Availability = Uptime / (Uptime + Downtime) Common Availability Tiers 95% - 18.26 days down per year 99% 99.9% 99.95% 99.99% 99.999% 5.26 minutes down per year

Achieve High Availability

- floating/elatic ip - move ip to another server
- multi - az -> multiple data centers . Multi o suporta nativ altfel ar fi greu sa o menti
    - cluster aware routing(load balancer dns) -> health checks
    - replication -> sync data
    - packet mirroring -> duplicate packets
- multi region -> multiple data centers in different regions
- minimizing impact radius -> split services
- proper proccesses -> human error
- monitoring -> alerts (db, web,vn,storage,vm)

Resilience

- ability of a symtem to remain functional when parts of it fail Resilient system is: ● Adaptive ● Self healing ● Predictable Requires ● Investment ● Automation ● Monitoring ● Simplicity

Disaster Recovery - DR

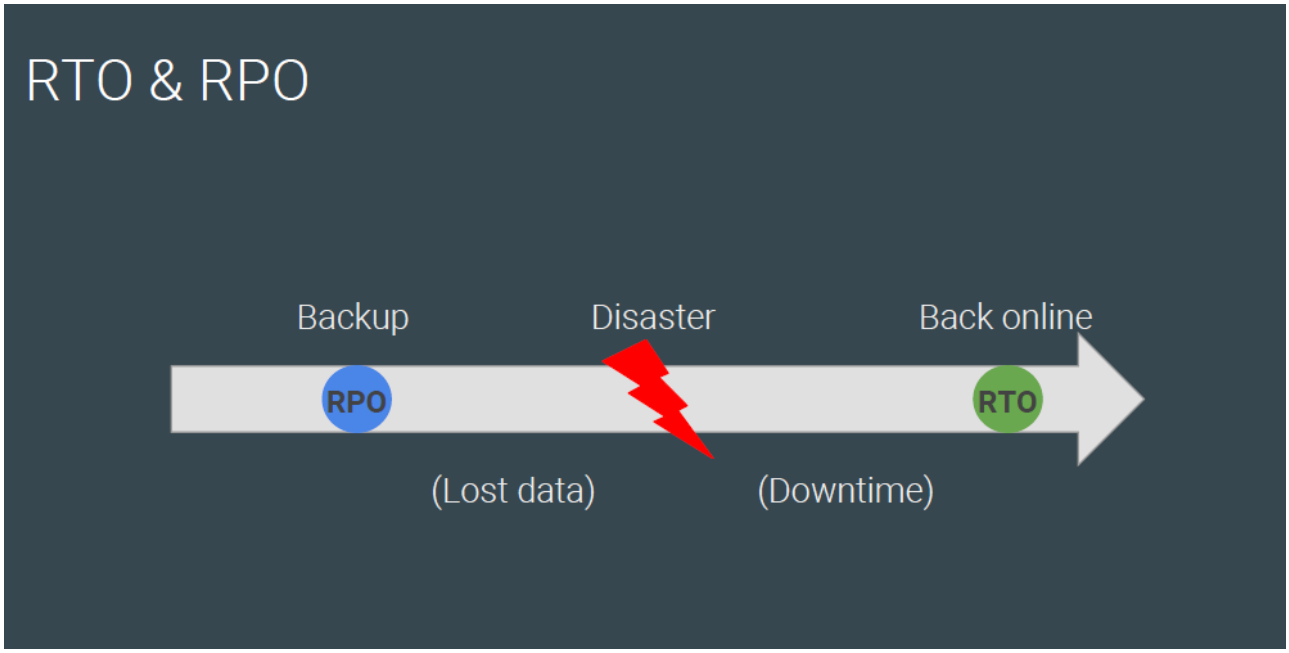
RTO

```
- cat dureaza
- depinde de DR strategy
- lower RTO - increase cost
```

RPO

- cata data e pierduta

- improve: frequent backups, replication, snapshots



DR Strategies

(in functie de la worse RTO la better dar mai scump)

1. Backup and Restore

- recreate everything based on backup

2. Pilot Light

- critical components pregatite
- copie a db ready for dr fara computare

3. Warm Standby

- full system replica ready, reduced size

4. Hot Site

- full replica ready